Adaptive Precision Dynamics in Deep Active Inference: From Lane Keeping to Obstacle Avoidance via Expected Free Energy Minimization

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Abstract

We present a deep active inference agent where adaptive precision-weighting enables lane keeping, obstacle avoidance, and post-evasion centering to emerge from a single Expected Free Energy (EFE) objective and shared RSSM world model. By dynamically modulating EFE component weights based on sensory prediction errors, the agent achieves 81.2% route completion on curves and 100% obstacle avoidance (28/28) in CARLA—without separate reward functions or control modules. We further investigate whether obstacle detection can be learned from the latent space alone, discovering a **posterior-prior epistemic gap**: an auxiliary prediction head achieves perfect obstacle encoding in the posterior (276 nats separation), yet avoidance remains at 0% because the RSSM transition model cannot propagate obstacle information through action-conditioned imagination.

1. Introduction

Active inference (Friston, 2010; Parr et al., 2022) unifies perception, learning, and action under a single variational principle: free energy minimization. An agent maintains a generative model of its environment and selects actions that minimize Expected Free Energy (EFE), naturally balancing goal-seeking (instrumental value) with information-seeking (epistemic value). This eliminates the need for separate reward functions and exploration strategies that characterize reinforcement learning approaches.

A key theoretical construct in active inference is *precision*—the inverse variance of prediction errors—which modulates the relative influence of different information channels. In biological systems, precision-weighting enables flexible behavior: the same neural circuitry produces different actions by adjusting which prediction errors are attended to (Parr et al., 2022). We ask: *can adaptive precision dynamics enable a single deep active inference agent to exhibit multiple driving behaviors?*

We demonstrate that the answer is yes. Using an RSSM world model (Hafner et al., 2023) with iCEM planning (Pinneri et al., 2021) in CARLA (Dosovitskiy et al., 2017), a single EFE objective with dynamically adjusted precision weights produces lane keeping, obstacle avoidance, and post-evasion lane return. Our contributions:

1. A **multi-component EFE formulation** combining latent-space instrumental value (GMM preference), ensemble-based epistemic value, decoded state-space preferences, and runtime obstacle penalties in a unified scoring function.
2. An **adaptive precision-weighting mechanism** where sensory prediction errors and environmental context dynamically modulate EFE component weights, enabling distinct behaviors to emerge from the same world model.
3. **Post-evasion centering** via temporary precision shifts that produce multi-maneuver obstacle avoidance sequences (evasion $\rightarrow$ lane return $\rightarrow$ evasion).
4. A systematic **investigation of vision-based obstacle detection** through latent representations, revealing a posterior-prior epistemic gap in the RSSM transition model.

2. Related Work

Deep active inference. Catal et al. (2020) and Fountas et al. (2020) apply deep active inference with VAE world models to visual navigation and Atari games. Millidge (2020) connects active inference with variational policy gradients, and Lanillos et al. (2021) surveys active inference in robotics. Delavari et al. (2025) demonstrate active inference for lateral vehicle control via perceptual motor learning, achieving lane keeping with minimal training data. However, these works focus on single behaviors and do not address how a single EFE objective can produce *multiple* distinct behaviors through precision modulation. We extend beyond single-task active inference to adaptive precision dynamics that enable behavioral flexibility—lane keeping, obstacle avoidance, and post-evasion centering—from a shared world model.

World models for control. DreamerV3 (Hafner et al., 2023) uses RSSM imagination for RL-based policy learning, achieving strong results across diverse domains. MILE (Hu et al., 2022) learns driving world models for imitation. These systems use the world model for policy optimization but do not examine whether the RSSM's imagined rollouts faithfully represent safety-critical objects like obstacles. We show that RSSM imagination exhibits an epistemic blind spot for obstacle-level reasoning, a limitation that may affect these systems as well.

Epistemic uncertainty in planning. Chua et al. (2018) use ensemble disagreement for model-based RL, and Sekar et al. (2020) plan to explore via world model uncertainty. These approaches assume that epistemic signals available during observation transfer to planning. We demonstrate that this assumption fails for RSSM-based active inference: posterior epistemic competence does not guarantee prior (imagination) competence.

Precision in active inference. Precision-weighting is central to biological active inference (Parr et al., 2022), modulating attention and sensory gain. While theoretical accounts describe how precision enables flexible behavior, computational implementations typically use fixed weights. We provide an empirical demonstration of *adaptive* precision dynamics in a deep AIF agent, where runtime modulation of EFE component weights produces qualitatively different driving behaviors.

3. Background

3.1. Active Inference and EFE

Under active inference, an agent selects policies π that minimize Expected Free Energy:

$$G(\pi) = \sum_t \gamma^t \Big[\underbrace{\mathbb{E}_q[-\log p_{\text{pref}}(z_t)]}_{\text{instrumental}} - \underbrace{H[p(o_t|s_t)]}_{\text{epistemic}} \Big] \quad (1)$$

where the instrumental term drives the agent toward preferred states p_{pref} and the epistemic term promotes information-seeking. Precision-weighting modulates the relative influence of these terms based on their estimated reliability.

3.2. RSSM World Model

The RSSM maintains deterministic state $h_t \in \mathbb{R}^{256}$ (GRU) and stochastic state $z_t \in \mathbb{R}^{64}$ (Gaussian):

$$\text{Posterior:} \quad z_t \sim q(z_t|h_t, o_t) \qquad \text{(filtering)} \quad (2)$$
$$\text{Prior:} \quad \hat{z}_t \sim p(z_t|h_t) \qquad \text{(imagination)} \quad (3)$$

The concatenated feature $[h_t; z_t] \in \mathbb{R}^{320}$ serves as the latent state for decoding and planning. During planning, only the prior is available—this distinction becomes critical in Section 6.

4. Method

4.1. Architecture

The world model comprises a ConvEncoder (4-layer CNN processing 64×64 RGB + 4D proprioceptive state $[\text{speed}, \text{steer}, h_{\text{err}}, c_{\text{err}}] \rightarrow$ 256D embedding), RSSM (deter=256, stoch=64), ObsDecoder (image reconstruction), StateDecoder (4D state prediction), and 5-head EnsembleTransitionHeads (epistemic uncertainty). Training minimizes Variational Free Energy (VFE) with dual KL divergence (Hafner et al., 2023), image and state reconstruction losses, on 96K frames of expert driving from CARLA Towns 04 and 06.

4.2. Multi-Component EFE

Our EFE scoring evaluates imagined trajectories from iCEM (500 samples, 50 elites, 5 iterations, horizon 12–15) with four components:

$$G(\tau) = \sum_{t=1}^{H} \gamma^t \Big[\beta_i G_i(t) - \beta_e G_e(t) + \beta_s G_s(t) + \beta_o G_o(t) \Big] \quad (4)$$

Instrumental value G_i: Cross-entropy $\mathbb{E}_q[-\log p_{\text{pref}}(z_t)]$ between imagined latents and a GMM preference fitted on expert driving latents (K=5, 64D). Z-scored across the CEM population per timestep.

Epistemic value G_e: Ensemble disagreement among 5 transition heads. Promotes information-seeking (negative sign in Eq. 4).

State-space value G_s: Decoded heading error and crosstrack error from the state decoder, penalizing deviation from lane center: $G_s(t) = \min(\hat{h}_t^2 + \hat{c}_t^2, 4.0)$. Not z-scored—state decoder predictions during imagination are noisy, and z-scoring amplifies this noise.

Obstacle penalty G_o: Runtime lane-change incentive using detected obstacle proximity and decoded crosstrack shift. Penalizes imagined trajectories that maintain the current lateral position when an obstacle is ahead.

Table 1. Task-specific precision configurations. Same world model checkpoint; only β weights and horizon differ.

Parameter	Task A	Task B
$\beta_{\text{instrumental}}$	1.0	1.0
$\beta_{\text{epistemic}}$	0.1	0.1
β_{state}	0.5	0.0
β_{obstacle}	0.0	40.0
Horizon H	12	15

Table 2. Obstacle avoidance results. SR: success rate. LC: mean lane changes per episode.

Route	SR	Avoided	MLD	LC
R0 (373m, 2 obs)	100%	10/10	0.67	6.0
R1 (518m, 3 obs)	100%	9/9	0.66	6.0
R2 (518m, 3 alt.)	100%	9/9	0.49	6.0
Total	**100%**	**28/28**	0.60	6.0

4.3. Adaptive Precision Dynamics

The core mechanism: β weights are modulated at runtime based on prediction errors and context, enabling different behaviors from the same EFE.

State error modulation. When observed state error $e = h_{\text{err}}^2 + c_{\text{err}}^2$ exceeds 0.3, the agent increases β_s (trust direct state observation) and decreases β_i (distrust GMM preference, which is out-of-distribution for off-center states):

$$\beta_s \leftarrow \beta_s \cdot \min(4, 1 + 4(e - 0.3)) \tag{5}$$

$$\beta_i \leftarrow \beta_i \,/\, \min(4, 1 + 4(e - 0.3)) \tag{6}$$

Obstacle proximity modulation. When an obstacle is detected ($\rho > 0.1$), the instrumental weight is suppressed to let the obstacle penalty dominate: $\beta_i \leftarrow \beta_i \cdot \max(0.2, 1 - 0.8\rho)$.

Post-evasion centering. After obstacle clearance, β_s is temporarily set to 0.5 with full heading+crosstrack penalty, pulling the ego back toward the route lane center. This deactivates when the ego returns to the route lane or a new obstacle is detected.

Table 1 shows the key difference: Task A uses β_s=0.5 for continuous centering while Task B sets β_s=0.0 to allow lateral deviation during lane changes, with β_o=40.0 for obstacle avoidance.

5. Emergent Driving Behaviors

All experiments use CARLA 0.9.16 with synchronous mode at 20 FPS.

5.1. Task A: Lane Keeping

On Town04 moderate curves (378m, 0–85 heading change), the agent achieves 81.2% average route completion with 0.63m mean lateral deviation (MLD). The state-space penalty (β_s=0.5) provides a Gaussian preference $\tilde{p}(s) = \mathcal{N}(0, I)$ over heading and crosstrack errors that keeps the agent centered through curves.

5.2. Task B: Obstacle Avoidance

On Town06 highway routes (373–518m, 2–3 static obstacles per route), the agent achieves 100% success rate with 28/28 obstacles avoided across 11 episodes on three route configurations (Table 2).

Each obstacle is avoided via an active lane-change maneuver (∼6 lane changes per episode), triggered by the obstacle proximity ramp modulating β_o and suppressing β_i.

5.3. Post-Evasion Centering

Route 2 places obstacles in alternating lanes. Without centering, the ego remains offset after the first evasion ($Y \approx 45$m) and passively clears subsequent obstacles. With the precision shift ($\beta_s \to 0.5$ after lock release), the ego returns toward the route lane between obstacles, requiring independent evasion decisions for each obstacle. This produces 6 active lane changes per episode rather than a single maneuver followed by passive clearance.

5.4. Shared World Model

Both tasks use the same trained world model checkpoint. The behavioral difference emerges entirely from precision weights: Task A emphasizes state-space centering (β_s=0.5, β_o=0), Task B emphasizes obstacle avoidance (β_s=0, β_o=40), and post-evasion centering temporarily blends both. This demonstrates that adaptive precision dynamics—not separate models or reward functions—can orchestrate multiple behaviors within a unified active inference framework.

6. Toward Vision-Based Obstacle Detection

The obstacle avoidance results above rely on runtime obstacle coordinates. We investigate whether this information can emerge from the latent space, removing all six privileged mechanisms (proximity computation, evasion direction, reflexive steer override, motor primitives, directional CEM penalty, obstacle EFE penalty).

6.1. Preference Gap Analysis

We fit a GMM on obstacle-free expert latents and measure the gap $\Delta = \log p(z_{\text{clean}}) - \log p(z_{\text{obs}})$. A positive gap would enable EFE-driven avoidance.

Table 3. Preference gap across modalities and methods.

Method	Δ	MLD	Avoid
Single GMM (different modalities)			
64×64 RGB	−24.6	0.31	0%
128×128 RGB	−2.3	0.17	0%
64×64 RGBD	−68.1	—	0%
Contrastive preference (α=0.2)			
Without aux head	+1.1	0.17	0%
With aux head	+133.1	0.61	0%

Δ: effective nats (positive = correct). Aux head: binary obstacle prediction during training only.

6.2. Contrastive Preference

We replace the single GMM with a log-ratio model: $\log p_{\text{pref}}(z) = \log p_{\text{clean}}(z) - \alpha \log p_{\text{avoid}}(z)$. This achieves correct posterior separation (37 nats raw at α=1.0, 1.1 effective at α=0.2), but avoidance remains 0%.

6.3. Auxiliary Prediction Head

To determine whether the bottleneck is the representation or the transition model, we add a supervised obstacle prediction head (MLP: 320→128→1, BCE loss, β_{aux}=1.0) during training only. This achieves **perfect** posterior separation (0.000 vs 1.000) and amplifies the contrastive gap to 276 nats (133.1 effective). Yet avoidance remains at **0%**.

6.4. The Posterior-Prior Epistemic Gap

These results isolate the blind spot to the RSSM **transition model** $f(h_t, z_t, a_t) \rightarrow h_{t+1}$. The posterior (filtering) encodes obstacle presence with perfect fidelity, but the prior (imagination) cannot predict how obstacles evolve as a function of steering actions. During CEM planning, all imagined trajectories—straight, left, right—produce similar latent sequences regardless of the posterior's obstacle encoding. The transition model has not learned obstacle-conditioned dynamics.

7. Discussion

Precision dynamics as a unifying mechanism. Our results demonstrate that adaptive precision-weighting is sufficient to produce multiple driving behaviors from a single EFE objective and world model. Lane keeping, obstacle avoidance, and post-evasion centering emerge from the interplay between four EFE components modulated by sensory prediction errors. This aligns with the active inference account of biological behavior, where precision-weighting enables flexible responses without separate control modules (Parr et al., 2022).

Posterior $\neq$ prior epistemic competence. The vision-based experiments reveal a fundamental dissociation: the RSSM posterior correctly encodes obstacle presence (276 nats with auxiliary supervision), but the transition model does not propagate this through imagination. This has direct implications for EFE-based planning: the planning quality is bounded by the transition model's ability to imagine action consequences, not by the representation's expressiveness. This finding may generalize to other RSSM-based systems (e.g., DreamerV3 (Hafner et al., 2023)) when applied to safety-critical scenarios requiring object-level reasoning.

Limitations. Obstacle detection currently relies on runtime coordinates rather than learned perception—a form of privileged information. While the precision-weighting framework is principled, the obstacle penalty G_o requires external obstacle positions. Sharp curves beyond ∼85 exceed the training distribution. All evaluation is on highway segments without junctions or traffic.

Future directions. Closing the posterior-prior gap requires transition models that learn object-conditioned dynamics. Attention-based transitions that selectively propagate object features through imagination, or object-centric world models (SLATE (Singh et al., 2022), SAVi (Kipf et al., 2022)) that decompose scenes into slots, may enable the transition model to predict how obstacles move through the visual field as a function of steering actions.

8. Conclusion

We demonstrated that adaptive precision dynamics enable a single deep active inference agent to exhibit multiple driving behaviors—lane keeping, obstacle avoidance, and post-evasion centering—through Expected Free Energy minimization with a shared RSSM world model. The same EFE objective, with dynamically adjusted β weights based on state prediction errors and obstacle proximity, produces 100% obstacle avoidance (28/28) and 81.2% lane-keeping completion without separate reward functions or control modules.

Our investigation into vision-based obstacle detection reveals a posterior-prior epistemic gap: the RSSM posterior encodes obstacles with perfect fidelity (276 nats via auxiliary supervision), but the transition model cannot imagine obstacle-conditioned dynamics. This dissociation—where posterior epistemic competence does not transfer to planning-level competence—identifies a critical frontier for active inference in safety-relevant domains.

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